

<u>S.No</u>	Barrier	Category
1	High Chip Fabrication & Initial Setup Cost	Resource Barriers
2	Inadequate Semiconductor Manufacturing & Testing Infrastructure	Resource Barriers
3	Inconsistent Power and Water Supply	Resource Barriers
4	Talent Drain & Skill Gaps in Semiconductor Industry	Resource Barriers
5	Limited R&D and Innovation Ecosystem	Resource Barriers
6	Slow Pace of Indian Semiconductor Mission (ISM & SPECS)	Regulatory Compliance Barriers
7	Complex Approval Processes & Limited Industrial Collaboration	Regulatory Compliance Barriers
8	Stringent and Overlapping Regulatory Norms	Regulatory Compliance Barriers
9	Globally Fragmented Semiconductor Supply Chain	Supply Chain & Operational Barriers
10	Inadequate & Fragile Domestic Ecosystem	Supply Chain & Operational Barriers
11	Lack of Value Chain Integration	Supply Chain & Operational Barriers
12	Lack of Specific & Sophisticated Transportation Network	Supply Chain & Operational Barriers
13	Limited Access to Critical Raw Materials & Rare Earth Elements	Supply Chain & Operational Barriers
14	Geopolitical Rivalry & Semiconductor Dominance	Global Conflicts & Trade Barriers
15	US-China Trade Conflicts & Restrictions	Global Conflicts & Trade Barriers
16	Taiwan Dominance & Alliances	Global Conflicts & Trade Barriers
17	Global Currency Volatility & Pricing Instability	Global Conflicts & Trade Barriers
18	IP Theft & Cybersecurity Threats Hindering Innovation	Innovation & Sustainability Barriers
19	Limited Semiconductor Logic Design Capabilities	Innovation & Sustainability Barriers
20	Lack of Circular Economy Models in Chip Manufacturing	Innovation & Sustainability Barriers
21	Water Regulations & Recycling Challenges (SDGs)	Innovation & Sustainability Barriers
22	Sanctions on AI & Quantum Chip Technologies	Global Conflicts & Trade Barriers
23	Unclear Standards for IP Protection in Semiconductor Supply Chain	Regulatory Compliance Barriers
24	Techno-nationalism & Strategic Stockpiling	Global Conflicts & Trade Barriers
25	Energy-Intensive Processes Clash with Net-Zero Goals	Innovation & Sustainability Barriers